INCH-POUND
MIL-I-24768/16
25 February 1992
SUPERSEDING
MIL-P-15035C
10 January 1962
MIL-P-79C
15 June 1961

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, COTTON-FABRIC-BASE, PHENOLIC-RESIN (FBM)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a woven cotton fabric substrate impregnated and bonded with a phenolic-resin matrix and processed to meet the requirements of this specification.

Intended use: The laminated material covered by this specification is suitable for gears and other applications requiring high impact strength and resiliency. It has excellent mechanical properties. It is not recommended for use in electrical applications.

Forms: Sheets, rolled tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 85, and an electrical strength test index of 85 for thicknesses up to 0.062 inch, and a mechanical strength test index of 125 and an electrical strength test index of 115 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)[3]
M24768/16-S-1	0.031	0.0065
M24768/16-S-2	0.047	0.0075
M24768/16-S-3	0.062	0.0075
M24768/16-S-4	0.094	0.009
M24768/16-S-5	0.125	0.010
M24768/16-S-6	0.156	0.011
M24768/16-S-7	0.188	0.0125
M24768/16-S-8	0.219	0.014
M24768/16-S-9	0.250	0.030
M24768/16-S-10	0.312	0.035
M24768/16-S-11	0.375	0.040
M24768/16-S-12	0.438	0.044
M24768/16-S-13	0.500	0.048
M24768/16-S-14	0.625	0.053
M24768/16-S-15	0.750	0.058
M24768/16-S-16	0.875	0.062
M24768/16-S-17	1.000	0.065
M24768/16-S-18	1.125	0.069
M24768/16-S-19	1.250	0.073
M24768/16-S-20	1.375	0.077
M24768/16-S-21	1.500	0.081
M24768/16-S-22	1.625	0.085
M24768/16-S-23	1.750	0.089
M24768/16-S-24	1.875	0.093
M24768/16-S-25	2.000	0.097
M24768/16-S-26	2.125	0.105
M24768/16-S-27	2.500	0.113
M24768/16-S-28	2.750	0.121
M24768/16-S-29	3.000	0.130
M24768/16-S-30	3.500	0.146
M24768/16-S-31	4.000	0.163
M24768/16-S-32	4.500	0.179
M24768/16-S-33	5.000	0.190
M24768/16-S-34	5.500	0.210
M24768/16-S-35	6.000	0.230
M24768/16-S-36	6.500	0.240
M24768/16-S-37	7.000	0.260
M24768/16-S-38	7.500	0.280
M24768/16-S-39	8.000	0.290
M24768/16-S-40	8.500	0.310
M24768/16-S-41	9.000	0.320
M24768/16-S-42	9.500	0.340
M24768/16-S-43	10.000	0.360
M24768/16-S-44	—[1]	—[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/16-S-44-0.8125.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

[3]Permissible variations for sheets greater than 0.219 inch thick shall be plus inch only.

Diameter of rolled round tubes: The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled round tube with an inside diameter of 0.375 inch and outside diameter of 0.500 inch shall be: M24768/16-TRR-0.375/0.500. The wall thickness tolerances for rolled round tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.375	48.000	0.500	50.000	0.062	2.000	1/2

TABLE III. *Wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)
0.062 to 0.124	0.015
0.125 to 0.249	0.020
0.250 to 0.500	0.020

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table IV. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.250 inch shall be: M24768-16R-0.250.

TABLE IV. *Size and tolerances for rods.*

Diameter (inches)	Tolerances (± inch)
0.125 to 1.999	0.005
2.000 to 4.000	0.008

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Performance requirements: Performance requirements for laminated materials shall be as specified in tables V, VI, and VII.

The following performance requirements do not apply to this plastic laminate:

a. Dissipation factor
b. Dielectric constant
c. Surface resistance
d. Volume resistivity
e. Dielectric strength
f. Arc resistance
g. Flammability
h. Silicone content
i. Thermal endurance
j. Tracking resistance.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE V. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A	4.6.4	Min. kV	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	10.0
Dielectric constant	A D-24/23	4.6.6	Max.										
Dissipation factor	A D-24/23	4.6.6	Max.										
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2]													
Lengthwise:	E-48/50	4.6.11	Min. ft–lbs. per inch				2.10	2.10	2.10	2.10	2.10	2.10	2.10
Crosswise:							1.90	1.90	1.90	1.90	1.90	1.90	1.90
Flexural strength													
Lengthwise:	A	4.6.12	Min. psi	17,000	17,000	17,000	17,000	17,000	17,000	16,000	16,000	15,000	15,000
Crosswise:				16,000	16,000	16,000	16,000	16,000	16,000	15,000	15,000	14,000	14,000
Thermal endurance													
Flexural strength:		4.6.13	Min. psi										

See footnotes at end of table.

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TABLE V. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Bonding strength[2]	A D-48/50	4.6.15	Min. pounds							1,800 1,600	1,800 1,600	1,800 1,600	1,800 1,600
Water absorption	D_1-24/23	4.6.17	Max. percent	8.00	4.40	3.20	2.50	1.90	1.60	1.20	1.10	1.00	1.00
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

TABLE VI. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)							
					0.063 – 0.094	0.095 – 0.125	0.126 – 0.188	0.189 – 0.250	0.251 – 0.375	0.376 – 0.500	0.501 – 1.000	1.000 – 2.000
Dielectric strength	A	4.6.5	Min. volts per mil	0.375 to 8.000								
Compressive strength[1]	A	4.6.14	Min. psi	0.375 to 8.000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000
Specific gravity	A	4.6.16	Min.	0.375 to 8.000	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.375 to 8.000	5.0	3.6	3.0	2.3	1.8	1.3	1.2	0.9

[1]Values apply to inside diameters of 0.375 inch and over and to outside diameters of 2.000 inches and less.

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TABLE VII. *Performance requirements for rods.*

Requirement	Cond.	Test para.	Unit	Diameter (inches)			
				0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flexural strength	A	4.6.12	Min. psi	16,000	16,000	16,000	
Compressive strength	A	4.6.14	Min. psi	19,000	19,000	19,000	19,000
Specific gravity	A	4.6.16	Min.	1.28	1.28	1.28	1.28
Water absorption	D_1-24/23	4.6.17	Max. percent	2.5	2.0	2.0	1.5

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Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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